Team Work Distribution

Team Member	Responsibilities
Retaj Majed alotaibi	Project Idea & Justification: - Wrote the project idea section - Explained the system importance - Defined target users
Shamayil Khouj	Problem Statement: - Wrote the problem section - Identified scheduling issues - Explained impact on clinics
Elham Alzahrani	Scope of Work: - Defined In Scope / Out of Scope - Organized features and user roles - Clarified project boundaries
Sahar Hawsawi	Project Timeline & References: - Prepared timeline table - Organized milestones - Numbered references and formatted document

Planning Stage

1. Project Idea and Justification

QueueLite is a lightweight appointment and queue management system designed for small clinics, salons, and consultation centers. The system allows clients to book, modify, or cancel appointments, while receptionists manage schedules and managers monitor daily reports.

The idea was chosen because many small clinics still rely on manual booking methods such as notebooks or messaging apps, which often result in lost appointments, double bookings, and poor schedule organization. QueueLite provides a simple digital solution tailored for small businesses.

2. Problem Statement

Small clinics and service centers frequently experience scheduling issues due to manual systems. Common problems include double booking, missed appointments, last-minute cancellations without notice, and lack of performance tracking. These issues reduce productivity and negatively affect customer satisfaction.

QueueLite addresses these challenges by centralizing appointment management and sending automated email reminders.

3. Scope of Work

In Scope:

• User roles: Client, Receptionist, Manager

• Login and role-based access control

• Book, modify, and cancel appointments

• Appointment status management (Scheduled / Cancelled / Completed)

• Email reminders before appointments

• Daily reports (bookings, cancellations, attendance rate)

Out of Scope:

• Online payment integration

• SMS or WhatsApp notifications

• AI-based scheduling

• Multi-branch support

• Integration with hospital systems

4. Project Timeline (Week 1 – Week 9)

Week	Stage / Milestone
Week 1	Requirements & Analysis
Week 2	System Design (ERD + UI Wireframes)
Week 3	Database Implementation
Week 4	Authentication Module (Login & Roles)
Week 5	Appointment Module (Book / Modify / Cancel)
Week 6	Email Reminder Integration
Week 7	Reporting Module
Week 8	Testing & Debugging
Week 9	Final Documentation & Presentation

Analysis Stage

1.1 Functional Requirements Analysis

FR1: The system shall allow users to log in using a username and password.

FR2: The system shall allow clients to view available time slots and book an appointment.

FR3: The system shall allow clients to modify or cancel their appointments.

FR4: The system shall allow receptionists to update appointment status (Scheduled, Cancelled, Completed).

FR5: The system shall send automated email reminders before appointments.

FR6: The system shall generate daily reports for managers including total bookings, cancellations, and attendance rate.

1.2 Non-Functional Requirements

NFR1: The system shall respond within 2 seconds.

NFR2: The system shall ensure data security through user authentication and password protection.

NFR3: The system shall be available 99% during working hours.

NFR4: The system shall support at least 50 concurrent users.

NFR5: The system shall be compatible with mobile and desktop devices.

NFR6: The system shall provide an easy-to-use interface.

2. User Analysis: Target Users and Their Needs

2.1. Target Users and Their Needs

User Role	Description	Needs (Requirements)
Client	Individuals seeking services at small clinics, salons, or consultation centers.	• A simple and easy-to-use interface to book, modify, or cancel appointments . • The ability to manage their bookings without needing to call the center directly.
Receptionist	Staff members responsible for daily schedule organization.	• A clear digital view of all scheduled appointments to prevent double bookings and lost appointments . • Tools to update and manage appointment statuses (Scheduled / Cancelled / Completed) in real-time .
Manager	Business owners or supervisors monitoring performance.	• Access to daily reports including total bookings, cancellations, and attendance rates . • The ability to monitor system activity to improve business productivity and customer satisfaction .

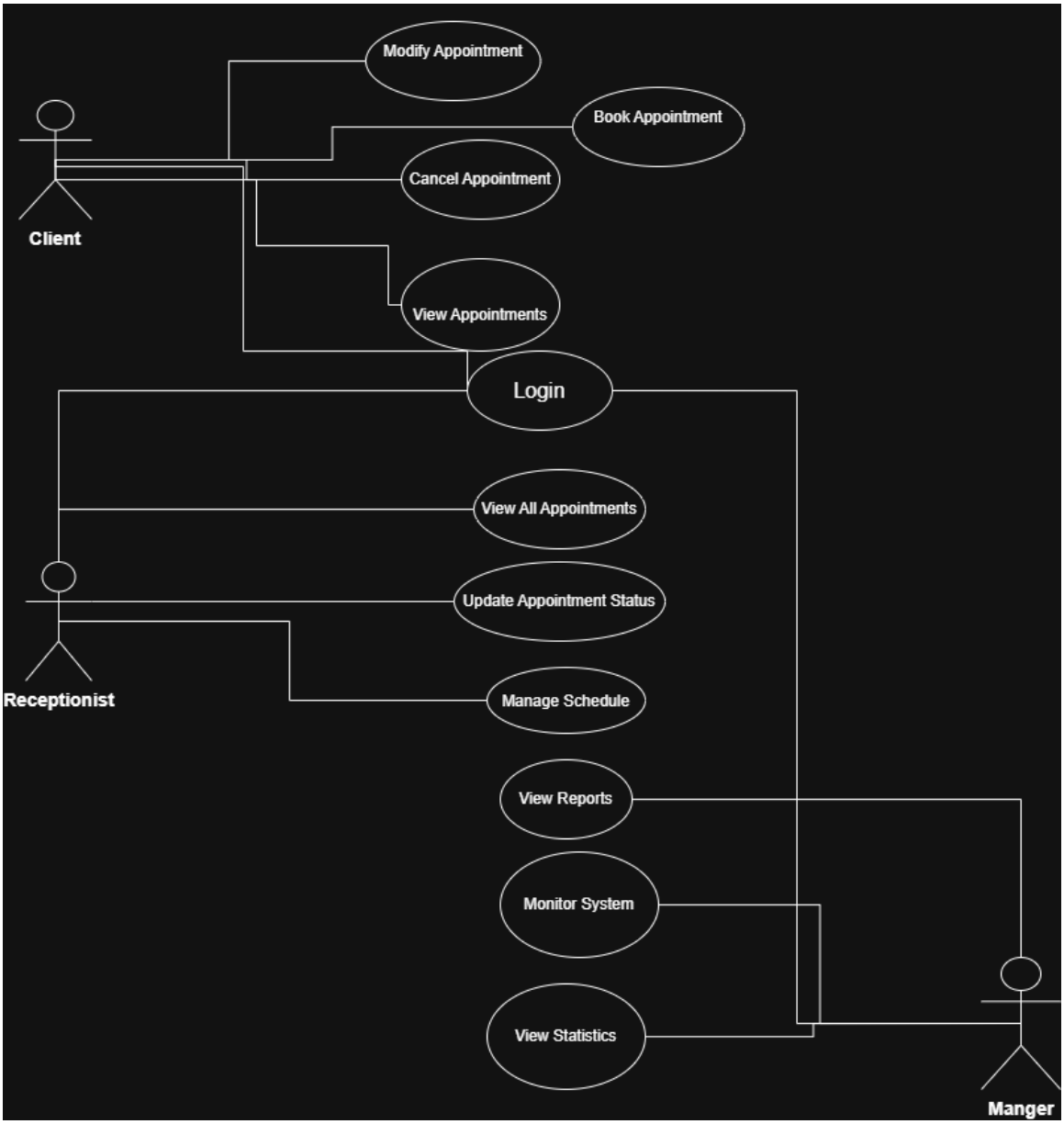
2.2. User Personas

Persona 1: The Client (Nora)

- Role: Client
- Background: A working professional with a tight schedule, who finds it inconvenient to make phone calls during business hours for appointments.
- Frustrations: Forgetting appointment details and long waiting times at clinics.
- Goals: To book, modify, or cancel appointments easily at any time via a user-friendly digital platform and receive automated email reminders.

Persona 2: The Receptionist (Ahmed)

- Role: Receptionist
- Background: Works at a busy medical center, managing schedules manually using notebooks and messaging apps.
- Frustrations: Double bookings, lost appointments, and the difficulty of tracking attendance rates.
- Goals: A centralized digital system to manage the daily schedule efficiently, prevent double bookings, and update appointment statuses (Scheduled / Cancelled / Completed) in real-time**.**



3. Potential challenges and risks : risk assessment and mitigation plan

Risk1 : double booking or scheduling conflicts

Impact : high - will effect the whole system reliability

Mitigation:

- Implement validation checks before confirming bookings
- Use database constraints to prevent overlapping appointments

Risk2 : Email reminder failure

Impact : medium – users may miss appointments

Mitigation :

- Add logs to track sent/failed emails
- Allow users to view upcoming appointments

Risk3: incorrect system usage

Impact : medium

Mitigation:

- Add input validation (e.g., required fields)
- Use clear labels and error messages
- Provide confirmation prompts before actions

Risk4 : Unauthorized Access

Impact : high

Mitigation**:**

  - Implement role-based access control
  - Encrypt passwords
  - Validate user sessions

Risk 5: Delays in Timeline

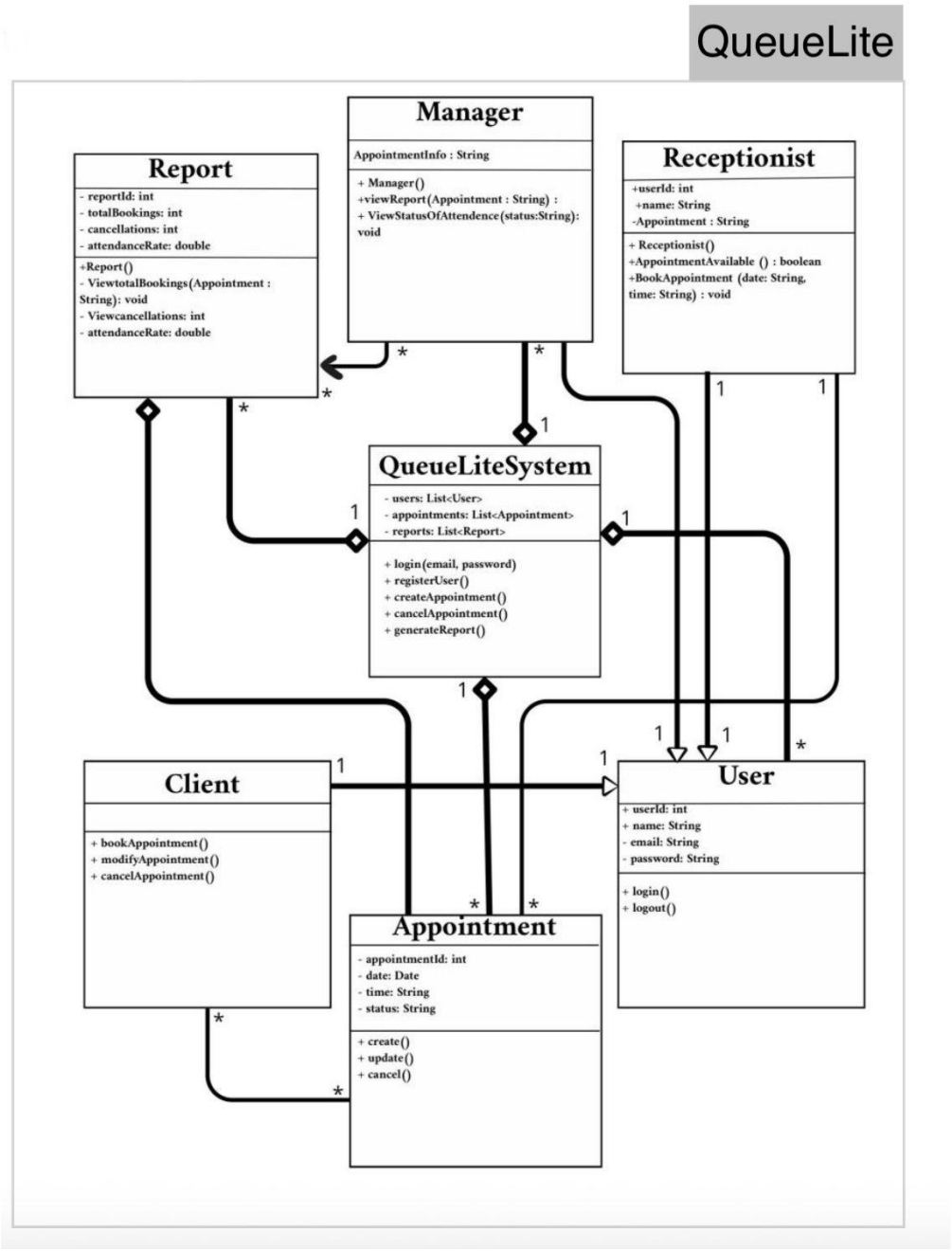
Impact : medium

Mitigation

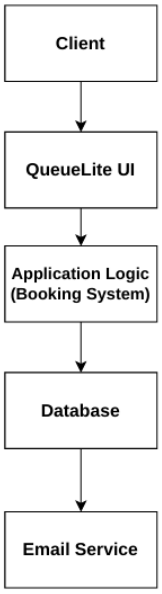
- Follow weekly milestones strictly
- Conduct weekly progress meetings
- Assign backup support

Design Stage

1.3 QueueLite UML Class diagram



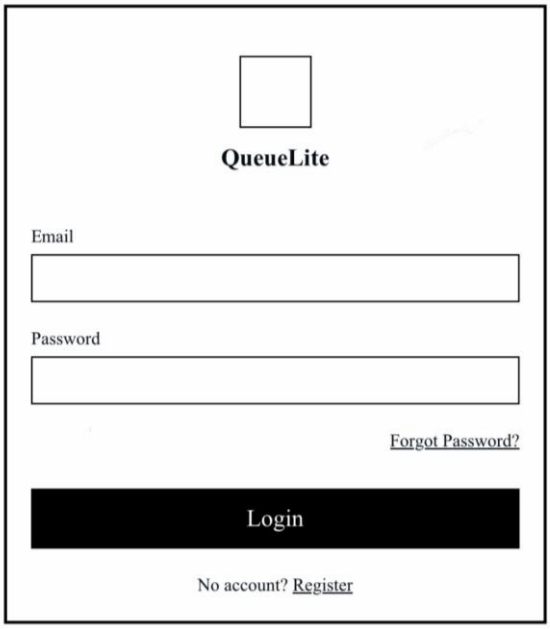
2.3 System Architecture Diagram



The architecture diagram illustrates how users interact with the QueueLite system through the user interface. The application logic processes booking operations and stores data in the database, while the email service sends automated reminders to users.

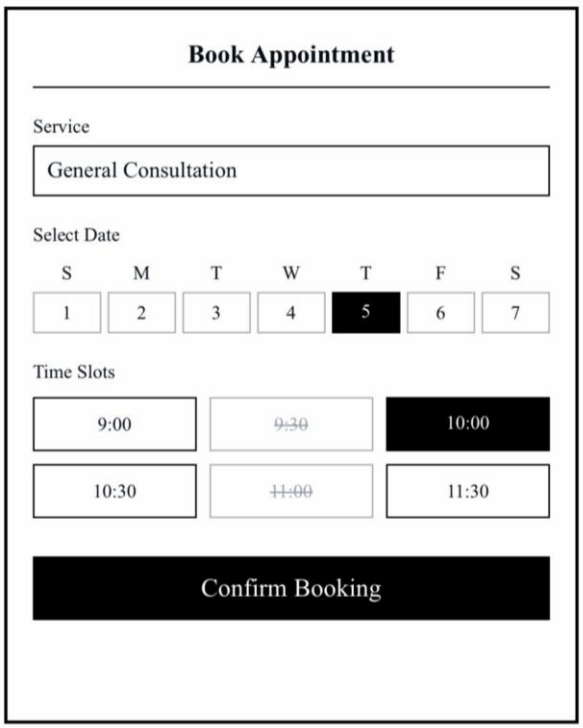
3.3 UI wireframes

1. Login Screen



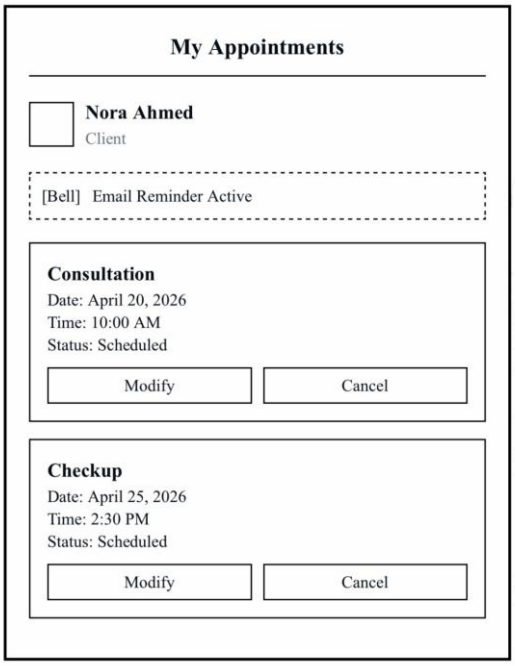
Description:
- Email input field with user icon
- Password input field with lock icon
- Forgot Password link for account recovery
- Large Login button for easy mobile access
- Register link for new users

2. Book Appointment



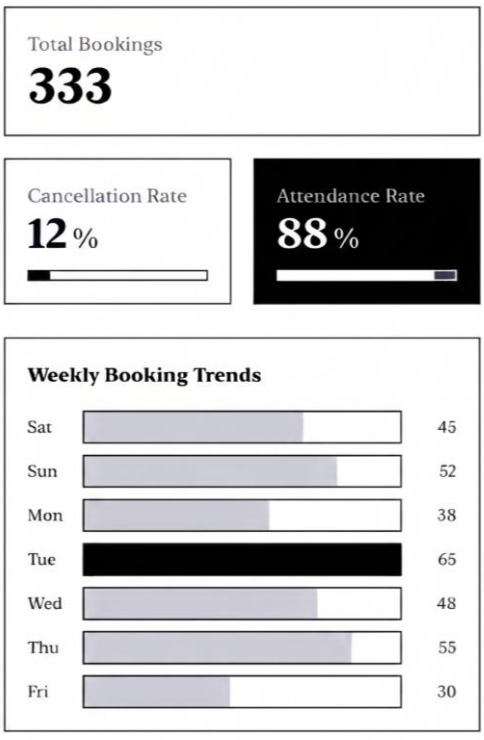
Description:
- Service selection dropdown
- Interactive calendar for date selection
- Available time slots grid (unavailable slots shown with strikethrough)
- Selected appointment summary
- Confirm Booking button

3. My Appointments (Nora)



Description:
- User profile section showing Nora Ahmed
- Email Reminder Active notification banner
- List of upcoming appointments with date, time, and status
- Modify button to reschedule appointments
- Cancel button to remove appointments

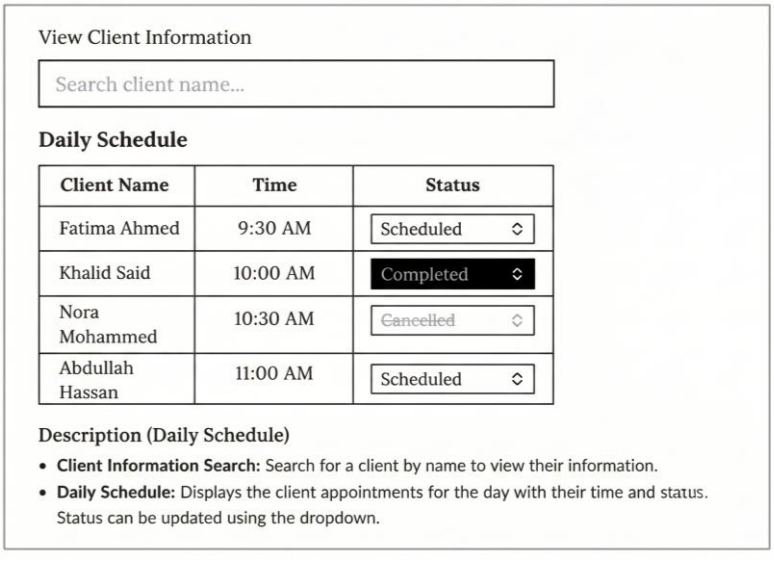
4- Daily Report



Description (Dashboard Overview)

- **Total Bookings:** Displays the total number of bookings in the system.
- **Cancellation Rate:** Shows the percentage of bookings that were cancelled.
- **Attendance Rate:** Shows the percentage of clients who attended.
- **Weekly Booking Trends:** Shows the number of bookings for each day of the week

5- Receptionist



Description (Daily Schedule)

- **Client Information Search:** Search for a client by name to view their information.
- **Daily Schedule:** Displays the client appointments for the day with their time and status. Status can be updated using the dropdown.

Testing Stage

4.1 Test Scenarios

TS01: Client Books an Appointment

The system should allow a registered client to select an available time slot and confirm a booking successfully.

TS02: Receptionist Updates Appointment Status

The system should allow the receptionist to update the appointment status from “Scheduled” to “Completed” after the client attends.

4.2 Test Cases

Test Case 1

Test Case ID: TC01

Related Scenario: TS01 – Client Books an Appointment

Description:
Verify that the client can successfully book an appointment using an available time slot.

Steps:

1. **Open the QueueLite system.**
2. **Login as client.**
3. **View available appointment slots.**
4. **Select a suitable time slot.**
5. **Click the “Book Appointment” button.**

Expected Result:

The system should confirm the appointment booking successfully and display a confirmation message.
Actual Result
The appointment was booked successfully and a confirmation message appeared.

Status: Pass

Test Case 2
Test case ID :TC02
Related Scenario: TS02 – Receptionist Updates Appointment Status

Description:
Verify that the receptionist can update the appointment status from Scheduled to Completed.

Steps:

1. **Open the QueueLite system.**
2. **Login as receptionist.**
3. **Open the appointments list.**
4. **Select a scheduled appointment.**
5. **Change status to "Completed."**
6. **Save changes.**

Expected Result

The system should update the appointment status successfully and display "Completed."

Actual Result:
The appointment status was updated successfully to "Completed."

Status: Pass

4.3 Bug Report

Bug Report 1

Bug ID: B01

Title: System Allows Double Booking for the Same Time Slot

Description:
The system allows more than one client to book the same appointment time slot, causing scheduling conflicts.

Steps to Reproduce:

1. **Login as Client A.**
2. **Book an available appointment slot (example: 10:00 AM).**
3. **Login as Client B.**
4. **Select the same appointment slot.**
5. **Confirm the booking.**

Expected Result
The system should prevent the second booking attempt and display a message such as:
"This time slot is already booked."

Actual Result:

The system accepts both bookings for the same time slot.

Status: Open

Bug Report 2
Bug ID : B01

Title: System Allows Double Booking for the Same Time Slot**.**

Description: The system allows more than one client to book the same appointment time slot, causing scheduling conflicts.

Steps to Reproduce:

1. Login as Client A.

2. Book an available appointment slot (example: 10:00 AM).

3. Login as Client B.

4. Select the same appointment slot.

5. Confirm the booking.

Expected Result: The system should prevent the second booking attempt and display a message such as:

“This time slot is already booked.”

Actual Result: The system accepts both bookings for the same time slot.

Status : open

Bug Report 2

Bug ID : B02

Title: Appointment Status Can Be Updated Without Existing Booking

Description: The system allows the receptionist to change the appointment status to “Completed” even when no valid appointment exists or when the appointment has already been cancelled**.**

Steps to Reproduce:

1. Login as Receptionist.

2. Search for a cancelled or non-existing appointment.

3. Attempt to update the appointment status to “Completed”.

4. Click Save/Update.

Expected Result:

The system should reject the update and display an error message such as: “Invalid appointment. Status cannot be updated.”

Actual Result:

The system updates the status successfully without validation.

Status : Open

References

1. Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.

2. Pressman, R. (2014). Software Engineering: A Practitioner's Approach. McGraw-Hill.

3. Oracle. (n.d.). Java Documentation. https://docs.oracle.com

4. Python Software Foundation. (n.d.). Python Documentation. https://docs.python.org

5. MySQL. (n.d.). MySQL Documentation. https://dev.mysql.com/doc/

6. https://wireframe.cc/